# SOLITON TECHNOLOGIES APTITUDE QUESTION BANK

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“A Soliton is a wave that travels without dispersion. Our name reflects our goal of creating a 'built-to-last' company.”

# Prepare the following topics

- **Basics C (Pointers and Strings)**
- **Basics Physics**
- **Volume Problems**
- **Trigonometry**
- **Probability Sums for Cards**
- **Electric Circuits (Resistors in Series and Parallel)**
- **Ratio Problems**
- **Problems in Acceleration, Velocity, Force (F=ma) and gravity.**

# Permutation

- There are numbers from 1 to n, which are inverted such that it results in numbers from n to 1. If only two consecutive numbers are stampled at a time and are inverted, how many moves are required?
- Ans: $\frac{n(n-1)}{2}$ moves.

# Area

- Construct a rectangle whose perimeter is 400cm such that it has the maximum area?
- Ans: **This will be a square (a special kind of rectangle) of sides 100 cm each.**

# Numbers

If i start counting with the numbers 10, 11, 12....in the left hand from little finger to thumb and counting the next number from fore finger to little finger and then back from little finger, ring finger, middle finger, fore finger to thumb and so on....on which finger will i count the number 1099?

Little : 10 18 26 34....(A.p of a=10, d=8)
Ring: 11 17 19 25 27 33...(mixed a.p)
Middle:12 16 20 24 28 32...(A.p of a=12,d=4)
index:13, 15 21 23 29 31...(mixed a.p)
Thumb:14 22 30...(a.p of a=14,d=8)

odd numbers are counted in ring & index fingers,& even on rest..

1099 is odd so it's counted in either ring or index fingers

let's see where does the even numbers 1100 & 1098comes.,

following an a.p so their middle finger will count 1099

# Velocity

Consider a boy standing on a building that is 500m height and another boy standing on the ground. If the boy on the ground throws a ball to the boy standing on the building, with what vertical velocity must the ball be thrown such that it reaches the boy standing on the building. Neglect the height of the boys.

G ( 9.8) acting downward, it would deaccelerate the ball

so to make the ball reach the top it should have some minimum velocity to cover 500m, let it be u.

Since its minimum it wud not travel any further than 500m, so final velocity is Zero.

$V^2 = U^2 + 2(\text{ acceleration}) * (\text{ distance})$

so $0 = u^2 - 2 ( 9.8) * (500)$

$u^2 = 9800$

$u= 98.99$ m/s

# Time & Distance

A person on a motor cycle gains a speed of 10kmph to 60kmph in 10 secs. The angular rotation of the needle in the speedometer is about 200 degrees. What will be the angular speed in degrees/ sec?

The angular speed of the speedomenter needle while accelerating at that rate is:
w = (200/360)*(2 pi)rad/10 sec
= (5/9)*2*pi/10 = pi/9 radians/s

# Progression

What do the vampires do who are more than one day old? Retire?

A vampire enters a town and bites 2 person every night. The bitten 2 persons grows in to two big vampires and bites 4 persons the next night. These 4 vampires bites 6 people the next night and the 6 bites 8 and so on.......how many vampires will be there on the 5th night?

The "bite rate" seems to be decreasing.
Fist one vampire bites two people (2 each)
Then two bite 4 (2 each)
Then 4 bite 6 (1.5 each)
Then 6 bite 8 (1.33 each)
If this trend continues, 8 bite 10, (1.25 each) on the fifth night.

# Time & Distance

Two cars start from the same point and move towards a destination that is 250km apart. They move with the speed of 60kmph and 40kmph. At certain point of time during the race, both the cars interchange their speed such that the second car reach the destination 1 hour earlier than the first car. What is the time in secs at which they interchange their speeds?

**let a = travel time when they interchanged speeds (in hrs)**
**let b = travel time for the 1st car to complete the trip**
**then (b-1) = travel time for the 2nd car to complete the trip**

**60a + 40b = 250**

**40a + 60(b-1) = 250**

**a = 26/20 a = 1.3 hrs (Ans) b = 17.2/4**

**b = 4.3 hrs travel time for the 1st car to complete the trip**

# Time & Distance

This question was about a journey between two towns. During the start of the journey the odometer reads 39593 ( the reverse of the no is also the same) and at the end of the journey again the odometer reads a number similar to prior one....If the max speed is 90km/hr during the journey. what will be the average speed throughout the journey?

The next palindromic number will be 39693,39793,39893,39993 40004, 40104  and so on. It is not given clearly whether next palindromic number. In that case the distance travelled will be 100. again the Max speed does not provide clear information about the change of speeds. Hence data insufficient.

# Volume

- .A rectangle is of length 8 inches, breath 11 inches and thickness 2 inches. When it is shaped to a cylindrical rod with the diameter is 10m. What is the height of the cylinder?
- I Don't see how the rectangular parallelopiped (**not** a rectangle, which is a 2-dimensional figure) could be deformed into a cylinder of height 10, but assuming you mean to find the dimensions of a cylinder with equal volume,

  volume = 8*11*2 = 176 $in^3$ = 2884.112 $cm^3$

  volume of rod is $\pi r^2 h$, so since r = 5m = 500cm,
  250000$\pi$h = 2884.112
  h = .00367

# Probability

Consider two identical pack of cards A and B. When one card from A is taken and shuffled with the card B, the first top card of A is the Queen of hearts. What will be the probability that the top card of B to be King of hearts?

There are two cases to be considered.

**CASE 1** : King of Hearts is drawn from Pack A and shuffled with Pack B

Probability of drawing King of Hearts from Pack A = $\frac{1}{51}$ (as Queen of Hearts is not to be drawn)

Probability of having King of Hearts on the top of the Pack B = $\frac{2}{53}$

So total probability of case 1 = $\frac{1}{51} * \frac{2}{53} = \frac{2}{51*53}$

**CASE 2 :** King of Hearts is not drawn from Pack A

Probability of not drawing King of Hearts from Pack A = $\frac{50}{51}$ (as Queen of Hearts is not to be drawn)

Probability of having King of Hearts on the top of the Pack B = $\frac{1}{53}$

So total probability of case 2 = $\frac{50}{51} * \frac{1}{53} = \frac{50}{51*53}$

Now adding both the probability, the required probability is

$$= \frac{2}{51*53} + \frac{50}{51*53} = \frac{52}{51*53} = 0.0192378$$

# Numbers

Consider a cube such that the product of the three faces of the cube forms the vertex.. The sum of all vertex is 1001. Then what will be the sum of numbers in all the faces of the cube.

- Let the integers written on each face be a,b,c,d,e, and f. Now, set up an equation with the given information. (Treat a as the top of the cube and f as the bottom.)
- $abc + acd + ade + aeb + fbc + fcd + fde + deb = 1001$
- $a(bc + cd + de + eb) + f(bc + cd + de + eb) = 1001$
- $(a + f)(bc + cd + de + eb) = 1001$
- $(a + f)(c(b+d) + e(b+d)) = 1001$
- $(a + f)(c + e)(b + d) = 1001 = 7 \times 11 \times 13$
- Since each of a, b, c , d , e, and f are positive, it follows that each of the three terms in the product is 2 or greater. Since 7, 11 and 13 are all prime, it follows that one of the terms must be 7, another 11 and the last 13. The desired sum is $2(7 + 11 + 13) = 62$.

# Velocity

A rocket launched accelerates at 3.5m/s^2 in 5.90 secs and 2.98m/s^2 in the next 5.98 secs and then experiences a free fall. What time will the rocket be in air? Assume that the rocket is launched from the ground.

d1 = 0.5*3.5*5.9^2 = 60.9 m
V1=a*t = 3.5*5.9 = 20.65 m/s. = Velocity after 3.5 s.

d2 = Vo*t + 0.5a*t^2
d2 = 20.65*5.98 + 2.98*5.98^2 = 176.8 m.
V2 = a*t = 2.98*5.98 = 17.82 m/s. = Velocity @ 176.8 m.

d3 = (V^2-Vo^2)/2g
d3 = (0-17.82^2)/-19.6 = 16.2 m. = Free fall distance up.
t = (V-Vo)/g = (0-17.82)/-9.8 = 1.82 s.
= Time to reach max. Ht.

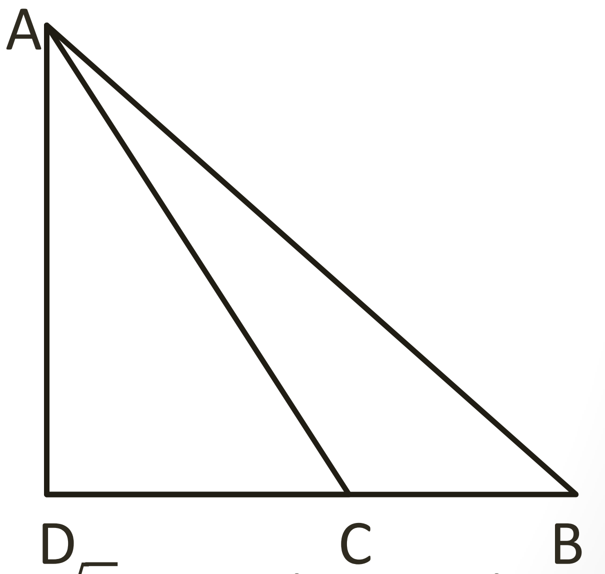
Tr = 5.90+5.98+1.82 = 13.7 s.
h = d1 + d2 + d3
h = 60.9 + 176.8 + 16.2 = 254 m. Above Gnd.

h = Vo*t + 0.5g*t^2 = 254 m.
0 + 4.9t^2 = 254
t^2 = 51.8
Tf = 7.2 s. = Fall time.

T = Tr + Tf = 13.7 + 7.2 = 20.9 s. = Time in air.

# Trigonometry

The angle of elevation to the tower is 30(degree) and then moved towards the tower of a distance 20m. Now the angle of elevation is 60(degree). What is the height of the tower?




Let AD be the tower.

$\angle ABD = 60^0$ and $\angle ACD = 30^0$. BC= 20m.

Let AD = x m and CD = y m.

Given tan $30^0 = \frac{x}{y+20} = \frac{1}{\sqrt{3}} \rightarrow 1.$

Tan $60^o = \frac{x}{y} = \sqrt{3} \rightarrow 2.$

From 1, $Y + 20 = x\sqrt{3}$.

From 2, $x = \sqrt{3}y = \sqrt{3}(x\sqrt{3} - 20) = 3x - 20\sqrt{3} = 3x - (20x1.73)$

$2x = 34.6$ m $\rightarrow x = 17.3$m.

# cOOL

A person likes to cool his small room during night so that he can sleep pleacefully but he has no air conditioner in his room. So he opens the door of the refrigerator that he has in his room. What will happen?

If you leave the door open, heat is merely recycled from the room into the refrigerator, then back into the room. A net room temperature increase would result from the heat of the motor that would be constantly running to move energy around in a circle.

# When we hit a foot ball, why does it goes along a curved path?

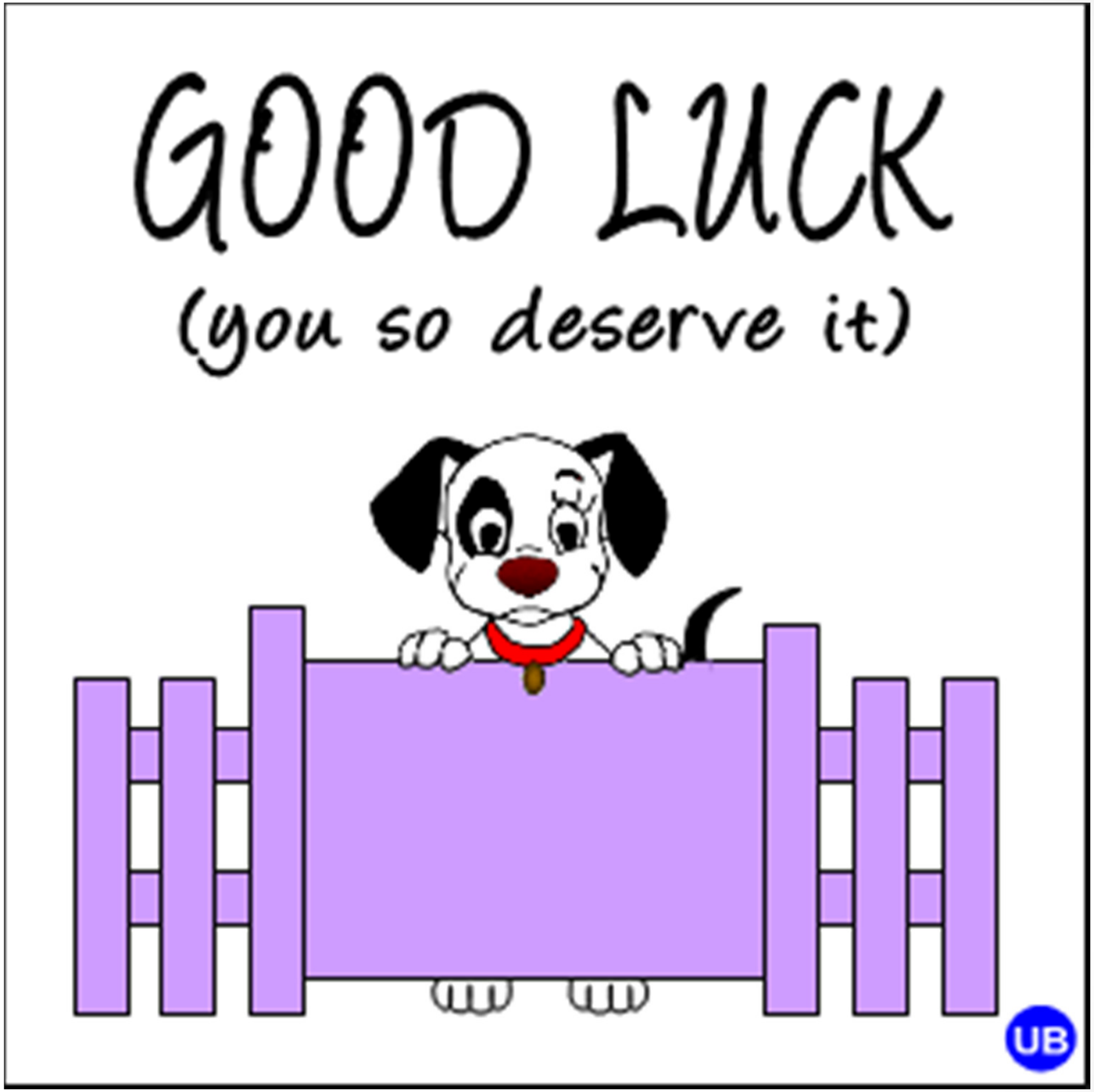
(we are not talking about the path of the projectile due to gravity).

Consider a ball that is spinning about an axis perpendicular to the flow of air across it. The air travels faster relative to the centre of the ball where the periphery of the ball is moving in the same direction as the airflow. This reduces the pressure, according to Bernouilli's principle. The opposite effect happens on the other side of the ball, where the air travels slower relative to the centre of the ball. There is therefore an imbalance in the forces and the ball deflects. This lateral deflection of a ball in flight is generally known as the "Magnus effect".

The forces on a spinning ball that is flying through the air are generally divided into two types: a lift force and a drag force. The lift force is the upwards or sidewards force that is responsible for the Magnus effect. The drag force acts in the opposite direction to the path of the ball.

The drag force, FD, on a ball increases with the square of the velocity, v, assuming that the density, r, of the ball and its cross-sectional area, A, remain unchanged: FD = CDrAv2/2. It appears, however, that the "drag coefficient", CD, also depends on the velocity of the ball. For example, if we plot the drag coefficient against Reynold's number - a non-dimensional parameter equal to rv D /µ, where D is the diameter of the ball and µ is the kinematic viscosity of the air - we find that the drag coefficient drops suddenly when the airflow at the surface of the ball changes from being smooth and laminar to being turbulent.

When the airflow is laminar and the drag coefficient is high, the boundary layer of air on the surface of the ball "separates" relatively early as it flows over the ball, producing vortices in its wake. However, when the airflow is turbulent, the boundary layer sticks to the ball for longer. This produces late separation and a small drag.


GOOD LUCK
(you so deserve it)
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